



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

TEST 1
SEMESTER III, SESSION 2020/2021

COURSE CODE : FSPC 0034
COURSE NAME : CHEMISTRY II
PROGRAMME : ASASI UTMSPACE
DURATION : 1 HOUR 15 MINUTES
DATE : 20 JANUARY 2021

INSTRUCTION TO CANDIDATES:

1. DO NOT OPEN THIS QUESTION PAPER UNTIL YOU ARE TOLD TO DO SO.
2. ANSWER ALL QUESTIONS.

WARNING!

Students caught copying/cheating during the examination will be liable for disciplinary actions and UTMSPACE may recommend the student to be expelled from the study.

This booklet consists of (6) printed pages including this page.

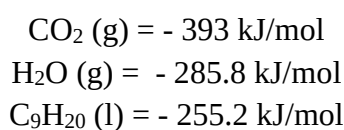
QUESTION 1 (10 MARKS)

- a) Determine whether each process below is an exothermic or endothermic reaction. Indicate the sign of ΔH for each process.

- i. Hand sanitizer evaporating from the skin. *endothermic*
- ii. Hot pack used as muscle pain relief. *exothermic*

(1+1 marks)

- b) Nonane is a colourless and flammable liquid. It is used as a solvent, distillation chaser and fuel additive. Given the ΔH of following compounds:



- i. Write the balanced chemical equation for the combustion of nonane, C_9H_{20} .



- ii. Determine the ΔH°_c of the reaction using the Hess's Law.

$$\Delta H^\circ_c = \Delta H_{\text{product}} - \Delta H_{\text{reactant}} = 9(-393) + 10(-285.8) - (-255.2) = -6139.8 \text{ kJ/mol}$$

- c) Gasoline, C_8H_{18} burns in oxygen to form nitrogen gas, CO_2 and water vapour, H_2O . Calculate the maximum temperature change when 4.62 g of liquid gasoline is burnt in a bomb calorimeter if heat capacity of the bomb calorimeter is 7.63 kJ/°C and ΔH°_c of the liquid gasoline is -5461 kJ/mol.

(4 marks)

$$m = 4.62 \text{ g}, \text{ m.m.} = 114 \text{ g/mol}$$

$$q = C_p \times \Delta T \quad \Delta H^\circ_c = -5461 = \frac{-q}{\text{mol}}$$

$$= 7.63(T)$$

$$\text{mol} = \frac{4.62}{114}$$

$$= 0.041$$

$$5461 = \frac{7.63 T}{0.041}$$

$$T = 30.31^\circ\text{C}$$

$$(a)(i) R_A = -\frac{d[A]}{dt} = -3.5 \times 10^{-5}$$

$$R = \frac{1}{9} \times \frac{d[A]}{dt} = \frac{1}{9} \times \frac{d[O]}{dt}$$

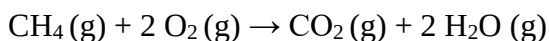
$$\frac{1}{9} \times 3.5 \times 10^{-5} = \frac{1}{2} \times \frac{d[O]}{dt}$$

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QUESTION 2 (10 MARKS)

$$3.5 \times 10^{-5} = \frac{1}{2} (R_{H_2O}), R_{H_2O} = 7.0 \times 10^{-5} \text{ M/s}$$

- a) Methane, CH₄ is an organic compound and one of the major constituents of natural gas, which is widely used as a fuel for ovens, homes and water heaters. Methane burns in oxygen at a reduction rate of $3.5 \times 10^{-5} \text{ M s}^{-1}$. Based on the following chemical equation:

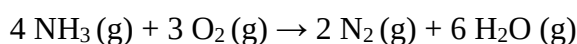


- Determine the formation rate of water vapour, H₂O.
- Calculate the time taken for methane, CH₄ to reduce from 0.1 M to 0.065 M.

$$-3.5 \times 10^{-5} = \frac{0.065 - 0.1}{t} \quad (1+2 \text{ marks})$$

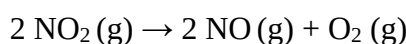
- b) Consider the following reaction at 550 K,

$$t = 1000 \text{ s}$$



Experiment	[NH ₃] (M)	[O ₂] (M)	Rate (M s ⁻¹)
1	0.25	0.15	5.0×10^{-5}
2	0.25	0.45	1.5×10^{-4}
3	0.50	0.15	2.0×10^{-4}

- Identify the overall order of reaction.
 - Determine the rate of reaction when [NH₃] = 0.10 M and [O₂] = 0.35 M
(2+2 marks)
- c) At high temperatures, nitrogen dioxide decomposes to nitric oxide and oxygen.



The reaction is an example of a second-order reaction and the rate constant was determined as $0.2 \text{ M}^{-1} \text{ s}^{-1}$.

- The initial concentration of NO₂ was 0.025 M. Determine the concentration of NO₂ after one hour.
- Calculate the time taken for the concentration of NO₂ to decrease to 20% of the initial concentration.

(2+1 marks)

$$(b) \frac{\text{Rate 2}}{\text{Rate 1}} = \frac{1.5 \times 10^{-4}}{5.0 \times 10^{-5}} = \frac{0.45^y}{0.15^y}, \quad 3 = 3^y, \quad y = 1$$

$$\frac{\text{Rate 3}}{\text{Rate 1}} = \frac{2.0 \times 10^{-4}}{5.0 \times 10^{-5}} = \frac{0.5^x}{0.25^x}, \quad 4 = 2^x, \quad x = 2$$

$$\text{Rate Law: Rate} = k[\text{NH}_3]^2[\text{O}_2]$$

$$\text{overall order of reaction} = 2 + 1 = 3$$

$$5.0 \times 10^{-5} = k(0.25)^2(0.15)$$

$$k = \frac{2}{375} \text{ M}^{-2}\text{s}^{-1}$$

$$R = \frac{2}{375} (0.10)^2 (0.35)$$

$$= 1.867 \times 10^{-5} \text{ M s}^{-1}$$

$$(c) k = 0.2, \quad 1 \text{ hour} = 60 \text{ min} = 3600 \text{ s}$$

$$R = k[\text{NO}_2]^2$$

$$\frac{1}{[A_t]} = kt + \frac{1}{[A_0]}$$

$$= 0.2(3600) + \frac{1}{0.025}$$

$$[A_t] = 1.32 \times 10^{-3} \text{ M}$$

$$[A_t] = 0.025 \times 0.2 = 5 \times 10^{-3} \text{ M}$$

$$\frac{1}{5 \times 10^{-3}} - \frac{1}{0.025} = 0.2(t)$$

$$160 = 0.2t$$

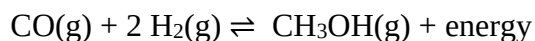
$$t = 800 \text{ s}$$

QUESTION 3 (10 MARKS)

- a) Explain how reaction rate and equilibrium are related concepts. Give an example to illustrate this relationship.

(2 marks)

- b) For the reaction,

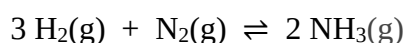


Predict the effect of the following changes on the equilibrium concentration of $\text{CH}_3\text{OH(g)}$:

- i. a decrease in pressure.
- ii. the addition of CO(g)

(1+1 marks)

- c) The Haber process involves the production of ammonia from hydrogen and nitrogen gas. In the laboratory, it was determined that the equilibrium concentrations of NH_3 , H_2 and N_2 are 0.003 M, 0.100 M, and 0.09 M, respectively.



- i. Write the equilibrium expression and calculate K_{eq} .
- ii. Describe the reaction progress when all three concentrations are at 3.0 M.
- iii. Predict the position of equilibrium.

(2+1+1 marks)

- d) Silver chromate (Ag_2CrO_4) is dissolved in 0.1 M AgNO_3 . Given that $K_{\text{sp}} = 9 \times 10^{-12}$.

- i. Write the reaction equation for the solubility of silver chromate.
- ii. Identify the effect of the common ion on the solubility.

(1+1 marks)

Appendix 1 : THE PERIODIC TABLE

[illegible]

Appendix 2 : LIST OF SELECTED CONSTANT VALUES

Speed of light in a vacuum	c	=	$3.0 \times 10^8 \text{ m s}^{-1}$
Avogadro's number	N_A	=	$6.02 \times 10^{23} \text{ mol}^{-1}$
Faraday constant	F	=	$9.65 \times 10^4 \text{ C mol}^{-1}$
Planck constant	h	=	$6.6256 \times 10^{-34} \text{ J s}$
Reduced Planck constant	$\hbar$	=	$1.054 \times 10^{-34} \text{ J s}$
Rydberg constant	R_H	=	$1.097 \times 10^7 \text{ m}^{-1}$ = $2.18 \times 10^{-14} \text{ J}$
Molar of gases constant	R	=	$8.314 \text{ J K}^{-1} \text{ mol}^{-1}$ = $0.08206 \text{ L atm mol}^{-1} \text{ K}^{-1}$
Boltzmann constant	k	=	$1.3807 \times 10^{-27} \text{ J K}^{-1}$
Mass of proton	M_p	=	$1.672 \times 10^{-27} \text{ kg}$
Electronic Bohr magneton	m_B	=	$9.2741 \times 10^{-24} \text{ J T}^{-1}$
Nuclear Bohr magneton	b_N	=	$5.05 \times 10^{-27} \text{ J T}^{-1}$
Vapour pressure of water	P_{water}	=	23.8 torr
Electron charge	e^-	=	$1.602 \times 10^{-19} \text{ C}$

UNIT AND CONVERSION FACTOR

Energy	$1 \text{ J} = 1 \text{ kg m}^2 \text{ s}^{-2} = 1 \text{ N m} = 10^7 \text{ erg}$ $1 \text{ calorie} = 4.184 \text{ Joule}$ $1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$ $1 \text{ amu} = 1.66 \times 10^{-27} \text{ kg}$
Pressure	$1 \text{ atm} = 760 \text{ mm Hg} = 760 \text{ torr} = 101.325 \text{ kPa} = 101325 \text{ N m}^{-2}$